

The equation we are solving is the following :

$$i \frac{\partial}{\partial t} \psi(x, t) = \left(-\frac{1}{2} \frac{\partial^2}{\partial x^2} + \frac{1}{2} \omega^2 x^2 \right) \psi(x, t), \quad \omega = 1 \quad (1)$$

with the fundamental as the initial condition, unless stated otherwise.

1 Analytical solution and convergence control

1.1 Time Independent Schrödinger Equation

We first solve the TISE given by

$$-\frac{1}{2} \frac{\partial^2}{\partial x^2} \varphi(x) + \frac{1}{2} \omega^2 x^2 \varphi(x) = E \varphi(x), \quad (2)$$

with the following boundary conditions

$$\varphi(x) \xrightarrow{x \rightarrow \pm \infty} 0. \quad (3)$$

The static solutions are given by the following equation,

$$\varphi_n(x) = \left(\frac{\omega}{\pi} \right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\sqrt{\omega} x) e^{-\omega x^2/2}, \quad (4)$$

whose the eigenvalues reads

$$E_n = \omega \left(n + \frac{1}{2} \right). \quad (5)$$

In the following, we set $\omega = 1$. The fundamental ($n = 0$) will be

$$\varphi_0(x) = (\pi)^{-1/4} e^{-x^2/2} \quad (6)$$

1.2 Time Dependent Schrödinger Equation

We want to solve (1), with the initial state given by (6). Since $H = -\frac{1}{2} \frac{\partial^2}{\partial x^2} + \frac{1}{2} x^2$ does not depend on time, the solution is simply given by

$$\psi(x, t) = \varphi_0(x) e^{-iHt} \quad (7)$$

$$= (\pi)^{-1/4} e^{-\frac{1}{2}(x^2 + it)} \quad (8)$$

Additionally, we can verify that the following general expression is a solution to (1)

$$\psi(x, t) = (\pi)^{-1/4} \exp \left[-\frac{1}{2} \left(x^2 + \frac{a^2}{2} (1 + e^{-2it}) + it - 2ax e^{-it} \right) \right], \quad (9)$$

where a can be any real parameter with the dimension of length. Let's note that for $a = 0$, we have the basic solution (8). We may rewrite this last general expression as the following

$$\psi(x, t) = (\pi)^{-1/4} \exp \left\{ -\frac{1}{2} \left[(x - a \cos t)^2 + i \left(t + 2ax \sin t - \frac{a^2}{2} \sin 2t \right) \right] \right\}. \quad (10)$$

1.3 TDSE for $V(x) = \omega^2(x+1)^2/2$

We want to solve the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(x,t) = \left(-\frac{1}{2}\frac{\partial^2}{\partial x^2} + \frac{1}{2}\omega^2(x+1)^2\right)\psi(x,t), \quad (11)$$

and the following initial condition

$$\Psi(x,0) = (\pi)^{-1/4} e^{-x^2/2}. \quad (12)$$

We can infer the solution from (1), by making the substitution

$$x \rightarrow y + 1 \quad (13)$$

We then have the following

$$dx = dy \quad \Longrightarrow \quad \frac{\partial}{\partial x} = \frac{\partial}{\partial y}, \quad (14)$$

$$x \rightarrow \infty \quad \Longrightarrow \quad y \rightarrow \infty, \quad (15)$$

and may rewrite the Schrödinger equation (1)

$$i\frac{\partial}{\partial t}\Psi(y,t) = \left(-\frac{1}{2}\frac{\partial^2}{\partial y^2} + \frac{1}{2}\omega^2(y+1)^2\right)\Psi(y,t), \quad \omega = 1. \quad (16)$$

The static solutions are still given by (4), that we may rewrite in terms of y and with $\omega = 1$, as

$$\varphi_n(y) = \left(\frac{1}{\pi}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(y+1) e^{-(y+1)^2/2}. \quad (17)$$

The solution to the time dependent problem is then given by (10), provided the substitution (13)

$$\psi(x,t) = (\pi)^{-1/4} \exp \left\{ -\frac{1}{2} \left[(x+1 - a \cos t)^2 + i \left(t + 2a(x+1) \sin t - \frac{a^2}{2} \sin 2t \right) \right] \right\}. \quad (18)$$

This is the most general solution to the Schrödinger equation (11). The initial condition (12) is satisfied by setting $a = 1$.

2 Propagators testing (Phys. Rev. E 96, 063307 (2017))

We want to solve the Schrödinger equation (11) by treating some of these terms as inhomogeneous terms, namely

$$i \frac{\partial}{\partial t} \psi(x, t) = \left[\left(-\frac{1}{2} \frac{\partial^2}{\partial x^2} + \frac{1}{2} x^2 \right) + \frac{1}{2} x + \frac{1}{2} (x + 1) \right] \psi(x, t), \quad \omega = 1 \quad (19)$$

$$= \left(H_0 + V_{\text{ext}} + h_I \right) \psi(x, t). \quad (20)$$

The general theory for ordinary differential equations consists in writing them as

$$dy = dt f(t, y), \quad y(t_0) = y_0, \quad (21)$$

To this end, we assess several methods provided by the above-mentioned paper to numerically solve equations of the type

$$\frac{dy}{dt} = Ly + N(y, t), \quad (22)$$

where L is the linear part and $N(y, t)$ is the nonlinear part. The equation we concern ourselves with is the time-dependent Kohn-Sham (TDKS) equation

$$i \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \mathbf{H} \Psi(\mathbf{r}, t), \quad (23)$$

$$= \mathbf{L} \Psi(\mathbf{r}, t) + \mathbf{N}(\Psi(\mathbf{r}, t)). \quad (24)$$

2.1 Analytical expressions

2.1.1 Time evolution operator approach in Schrödinger picture

A formal solution to (24) is

$$\Psi(\mathbf{r}, t) = U(t, t_0) \Psi(\mathbf{r}, t_0), \quad (25)$$

where $U(t, t_0)$ is the evolution operator defined as

$$U(t, t_0) = \mathcal{T} \exp \left\{ -i \int_{t_0}^t \mathbf{H}(\mathbf{r}, \tau) d\tau \right\}, \quad (26)$$

with $\mathcal{T}$ being the time-ordering operator. In practice, the above expression is split into a product of evolution operators as

$$U(t_n, t_0) = \prod_{k=1}^n U(t_k, t_{k-1}), \quad t_n = n\Delta t. \quad (27)$$

For sufficiently small steps, we assume the Hamiltonian is kept constant between the two instants and write

$$U(t_{n+1}, t_n) = \exp \left[-i \mathbf{H}(\mathbf{r}, t_n) \Delta t \right]. \quad (28)$$

2.1.2 Implicit-explicit (IMEX) schemes

The equation (21) can be straightforwardly and conveniently solved by approximating it, for sufficiently small steps $\Delta t \rightarrow 0$, to

$$y_{n+1} = y_n + \Delta t f(t_n, y_n) + O(\Delta t^2), \quad y(t_0) = y_0. \quad (29)$$

This expression known as an *explicit scheme* interestingly consider that, for sufficiently small enough steps, the function did not change much between t_n and t_{n+1} .

An alternative to (29) is the *implicit scheme* given by the following equation

$$y_{n+1} = y_n + \Delta t f(t_{n+1}, y_{n+1}) + O(\Delta t^2), \quad y(t_0) = y_0. \quad (30)$$

This last expression is preferable whenever the problem is said to be *stiff*, as it is much stabler.

2.1.3 Integrating factor (IF) method

The idea here is to use an integrating factor to simplify the resolution of the equation. As far as quantum mechanics is concerned, it is about using the *interaction picture* defined as the following

$$\Phi(\mathbf{r}, t) = e^{iLt} \Psi(\mathbf{r}, t). \quad (31)$$

Inserting this into (24) yields

$$i \frac{\partial}{\partial t} \Phi(\mathbf{r}, t) = e^{iLt} \mathbf{N} (e^{-iLt} \Phi, t). \quad (32)$$

Reversing back to $\Psi(\mathbf{r}, t)$ form (31) subsequently integrating from t_n to t_{n+1} yields

$$\Psi(\mathbf{r}, t_{n+1}) = e^{-iL\Delta t} \Psi(\mathbf{r}, t_n) - ie^{-iLt_{n+1}} \int_{t_n}^{t_{n+1}} e^{iL\tau} \mathbf{N} (e^{-iL\tau} \Phi, \tau) d\tau. \quad (33)$$

2.1.4 Exponential time differencing (ETD) method

From (24), we may express $\mathbf{N}(\Psi, t)$ as a sum of the linear part and the time-derivative. Subsequently using the identity

$$i \frac{\partial}{\partial t} e^{iLt} \Psi(\mathbf{r}, t) = e^{iLt} \left[-L(\mathbf{r}, t) \Psi(\mathbf{r}, t) + i \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) \right], \quad (34)$$

one can rewrite (24) as

$$i \frac{\partial}{\partial t} \Phi(\mathbf{r}, t) = e^{iLt} \mathbf{N} (\Psi, t), \quad (35)$$

Integrating from t_n to t_{n+1} yields

$$\Psi(\mathbf{r}, t_{n+1}) = e^{-iL\Delta t} \Psi(\mathbf{r}, t_n) - ie^{-iLt_{n+1}} \int_{t_n}^{t_{n+1}} e^{iL\tau} \mathbf{N} (\Psi, \tau) d\tau. \quad (36)$$

The expressions (35) and (36) are formally identical to (32), (33), safe the variable change in the treatment of the nonlinear term. In other words, the IF method consists in propagating the solution in the interaction picture whereas the ETD method does so in the Schrödinger picture.

From this last remark, there is an argument to make about the point of propagating the solution by the means of (33) or (36). Indeed, one can just propagate the solution in the interaction picture all the way through the last integration step

$$\Phi(\boldsymbol{r}, t_{n+1}) = \Phi(\boldsymbol{r}, t_n) - i \int_{t_n}^{t_{n+1}} e^{iL\tau} \boldsymbol{N} \left(e^{-iL\tau} \Phi, \tau \right) d\tau, \quad (37)$$

and then, and then only, obtain the solution $\Psi(\boldsymbol{r}, t_f)$ from $\Phi(\boldsymbol{r}, t_f)$ by making use of (31).

2.2 Solutions to the Time Dependent Density Functional Theory

In this part, we address some specific technics to solve the TDKS equation, either in the Schrödinger picture (25) or in the interaction picture (31).

2.2.1 Taylor time propagation

From the expression (25), a simple Taylor expansion yields

$$\Psi(\mathbf{r}, t_{n+1}) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-i\Delta t \mathbf{H}(t_n) \right)^k \Psi(\mathbf{r}, t_n) \quad (38)$$

The developpment is usually truncated after n terms, which in turns gives an n th-order approximation.

$$\Psi(\mathbf{r}, t_{n+1}) = \sum_{k=0}^n \frac{1}{k!} \left(-i\Delta t \mathbf{H}(t_n) \right)^k \Psi(\mathbf{r}, t_n) + O(\Delta t^{n+1}). \quad (39)$$

Most propagation schemes seems to be essentially just that in some way or another.

2.2.2 Split-operator method

2.2.3 Crank-Nicholson method

By making use of both the explicit and implicit schemes and averaging them, we achieve the Crank Nicholson propagation scheme given by the following formula

$$y_{n+1} = y_n + \frac{\Delta t}{2} \left(f(t_{n+1}, y_{n+1}) + f(t_n, y_n) \right) + O(\Delta t^3) \quad (40)$$

The application of this formula to the problem (24) gives

$$\Psi_{n+1} = \Psi_n - \frac{i\Delta t}{2} \left(\mathbf{H}_{n+1} \Psi_{n+1} + \mathbf{H}_n \Psi_n \right), \quad \mathbf{H}_i \equiv \mathbf{H}(\mathbf{r}, t_i) \quad (41)$$

$$\left\{ \mathbb{1} + \frac{i\Delta t}{2} (\mathbf{L}_{n+1} + \mathbf{N}_{n+1}) \right\} \Psi_{n+1} = \left\{ \mathbb{1} + \frac{i\Delta t}{2} (\mathbf{L}_n - \mathbf{N}_n) \right\} \Psi_n \quad (42)$$

2.2.4 Second-order IMEX method

2.2.5 Integrating factor method with explicit multistep

This method is not derived from (29) but rather from the *Adam-Bashfort* formula, which reads

$$y_{n+1} = y_n + \frac{\Delta t}{2} \left(3f(y_n, t_n) - f(y_{n-1}, t_{n-1}) \right) + O(\Delta t^3), \quad \Delta t = t_{n+1} - t_n. \quad (43)$$

The application of this formula to (31) gives

$$\Phi_{n+1} = \Phi_n + \frac{i\Delta t}{2} \left[-3e^{i\mathbf{L}t_n} \mathbf{N} (e^{-i\mathbf{L}t_n} \Phi_n, t_n) + e^{i\mathbf{L}t_{n-1}} \mathbf{N} (e^{-i\mathbf{L}t_{n-1}} \Phi_{n-1}, t_{n-1}) \right], \quad (44)$$

$$\Psi_{n+1} = e^{-i\mathbf{L}\Delta t} \Psi_n - \frac{3i\Delta t}{2} e^{-i\mathbf{L}\Delta t} \mathbf{N} (\Psi_n, t_n) + \frac{i\Delta t}{2} e^{-2i\mathbf{L}\Delta t} \mathbf{N} (\Psi_{n-1}, t_{n-1}). \quad (45)$$

2.2.6 Integrating factor with Runge-Kutta

Runge Kutta methods propagate the solution over an interval by combining the information from several Euler-style steps (each involving one evaluation of the right-hand function), and then using the information obtained to match a Taylor series expansion up to some order.

— Euler method (RK1)

The formula for the Euler method is given by

$$k_1 = f(t_n, y_n), \quad (46)$$

$$y_{n+1} = y_n + \Delta t k_1 + O(\Delta t^2). \quad (47)$$

The formula for the explicit Euler method (29) advances the solution one step forward in time. The method is described to be $O(\Delta t^2)$ or $o(\Delta t)$, thus of order 1. Though imprecise and unstable, Euler method may be used to obtain a partial solution that is then used in more advanced propagation schemes.

$$k_1 = -ie^{-iL\Delta t} \mathbf{N}(\Psi_n, t_n), \quad (48)$$

$$\Psi_{n+1} = e^{-iL\Delta t} \Psi_n + \Delta t k_1. \quad (49)$$

— Second-order Runge-Kutta methods

Using (46) as a trial step to some point of the interval and subsequently using the value of t and y at this point, we may compute the real step as

$$k_1 = f(y_n, t_n), \quad (50)$$

$$k_2 = f(t_n + \alpha\Delta t, y_n + \alpha\Delta t k_1), \quad (51)$$

$$y_{n+1} = y_n + \Delta t \left((1 - \beta)k_1 + \beta k_2 \right) + O(\Delta t^3), \quad (52)$$

where $\beta = (1 - \alpha)$ and α is some non-zero real parameter. For example $\alpha = 1/2$ is the second-order Runge's method also known as the *midpoint rule*. Other popular choices are $\alpha = 2/3$ which is Ralston's method and $\alpha = 1$ which is Heun's method.

Using (52) on the relation (31) yields

$$k_1 = -ie^{-iL\Delta t} \mathbf{N}(\Psi_n, t_n), \quad (53)$$

$$k_2 = -ie^{-i\alpha L\Delta t} \mathbf{N}(\Psi_n + \alpha\Delta t k_1, t_n + \alpha\Delta t), \quad (54)$$

$$\Psi_{n+1} = e^{-iL\Delta t} \Psi_n + \Delta t \left((1 - \beta)k_1 + \beta k_2 \right). \quad (55)$$

— Third-order Runge-Kutta methods

Following the second-order Runge-Kutta methods are the third-order methods described by the equations

$$k_1 = f(y_n, t_n), \quad (56)$$

$$k_2 = f(t_n + c_2\Delta t, y_n + a_{21}\Delta t k_1), \quad (57)$$

$$k_3 = f(t_n + c_3\Delta t, y_n + \Delta t (a_{31}k_1 + a_{32}k_2)), \quad (58)$$

$$y_{n+1} = y_n + \Delta t (b_1k_1 + b_2k_2 + b_3k_3) + O(\Delta t^4), \quad (59)$$

where the coefficients c_i , a_{ij} and b_j are given by the *Butcher tableau*.

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^T \end{array}$$

— **Fourth-order Runge-Kutta methods**

Following are the fourth-order Runge-Kutta methods described by the equations

$$k_1 = f(y_n, t_n), \quad (60)$$

$$k_2 = f(t_n + c_2 \Delta t, y_n + a_{21} \Delta t k_1), \quad (61)$$

$$k_3 = f(t_n + c_3 \Delta t, y_n + \Delta t (a_{31} k_1 + a_{32} k_2)), \quad (62)$$

$$k_4 = f(t_n + c_4 \Delta t, y_n + \Delta t (a_{41} k_1 + a_{42} k_2 + a_{43} k_3)), \quad (63)$$

$$y_{n+1} = y_n + \Delta t (b_1 k_1 + b_2 k_2 + b_3 k_3 + b_4 k_4) + O(\Delta t^5), \quad (64)$$

where the coefficients c_i , a_{ij} and b_j are, again, given by the *Butcher tableau*.

There exist two popular fourth-order Runge-Kutta methods, only differing by the coefficients of the Butcher tableau.

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & 1/6 & 1/3 & 1/3 & 1/6 \end{array}$$

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 0 & 0 & 0 \\ 2/3 & -1/3 & 1 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 \\ \hline & 1/8 & 3/8 & 3/8 & 1/8 \end{array}$$